Presentation Title

Personal Loan Campaign: Machine Learning Project

3.28.2026

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Executive Summary

- I developed a highly accurate model that identifies customers likely to accept a personal loan. The final model achieves perfect recall and near-perfect precision, meaning I capture all potential customers while minimizing wasted outreach.
- The biggest drivers are income, education, and spending behavior.

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Business Problem Overview and Solution Approach

Business Problem

- The bank aims to **increase personal loan conversion rates**
- Current marketing campaigns are **broad and inefficient**

This results in:

- High marketing costs
- Low conversion efficiency

Key Objectives:

- Identify customers with a **high likelihood of loan acceptance**
- Reduce **unnecessary marketing spend**
- Improve **overall return on investment (ROI)**

Solution Approach

- A structured machine learning pipeline was implemented:
- Exploratory Data Analysis (EDA)
- Data preprocessing and feature preparation
- Decision Tree model development
- Model optimization using pre-pruning
- Model refinement using post-pruning
- Performance evaluation and model selection

Approach Summary:

- Leveraged data-driven insights to build an interpretable model
- Applied pruning techniques to improve generalization and performance
- Selected the best model based on balanced evaluation metrics

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EDA Results: Descriptive Statistics

- **Data Overview and Summary Statistics**
- The dataset contains 5000 customer records with 14 variables describing demographic, financial, and behavioral characteristics.
- The target variable, Personal_Loan, shows that only 9.6% of customers accepted the loan, indicating a class imbalance problem. Therefore, evaluation metrics such as recall, precision, and F1-score will be more important than accuracy.
- Income and CCAvg exhibit right-skewed distributions, with a small number of high-value customers, suggesting that these variables may be strong predictors of loan acceptance.
- The Experience variable contains negative values, which are not logically valid and indicate data quality issues that will need to be addressed during preprocessing.
- Mortgage values are highly skewed, with many customers having no mortgage and a few having very high values.
- Binary variables such as CD_Account, Securities_Account, and Online usage represent customer engagement and are likely to play an important role in predicting loan acceptance.

	count	mean	std	min	25%	50%	75%	max
ID	5000.0	2500.500000	1443.520003	1.0	1250.75	2500.5	3750.25	5000.0
Age	5000.0	45.338400	11.463166	23.0	35.00	45.0	55.00	67.0
Experience	5000.0	20.104600	11.467954	-3.0	10.00	20.0	30.00	43.0
Income	5000.0	73.774200	46.033729	8.0	39.00	64.0	98.00	224.0
ZIPCode	5000.0	93169.257000	1759.455086	90005.0	91911.00	93437.0	94608.00	96651.0
Family	5000.0	2.396400	1.147663	1.0	1.00	2.0	3.00	4.0
CCAvg	5000.0	1.937938	1.747659	0.0	0.70	1.5	2.50	10.0
Education	5000.0	1.881000	0.839869	1.0	1.00	2.0	3.00	3.0
Mortgage	5000.0	56.498800	101.713802	0.0	0.00	0.0	101.00	635.0
Personal_Loan	5000.0	0.096000	0.294621	0.0	0.00	0.0	0.00	1.0
Securities_Account	5000.0	0.104400	0.305809	0.0	0.00	0.0	0.00	1.0
CD_Account	5000.0	0.060400	0.238250	0.0	0.00	0.0	0.00	1.0
Online	5000.0	0.596800	0.490589	0.0	0.00	1.0	1.00	1.0
CreditCard	5000.0	0.294000	0.455637	0.0	0.00	0.0	1.00	1.0

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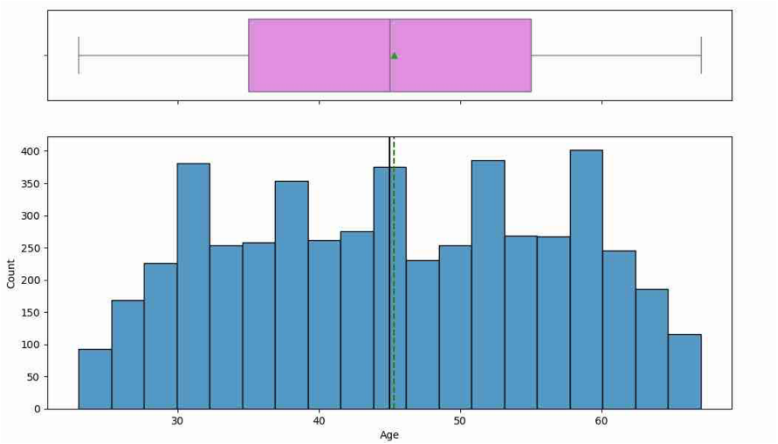
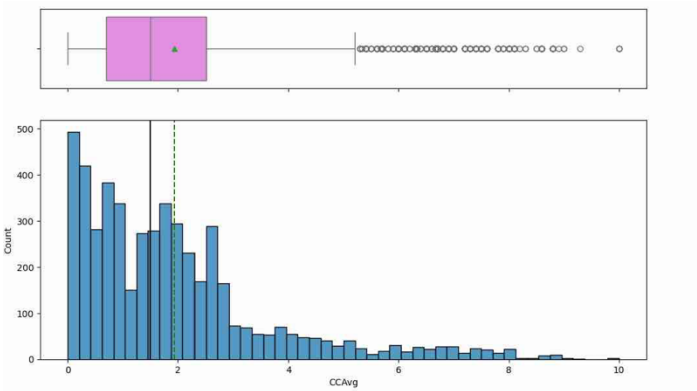
EDA Results: Univariate Analysis

Age
•Fairly **normally distributed**, centered around **45 years**.
•Range spans from early 20s to mid-60s.
Insight:
•The dataset represents a **balanced working-age population**.
Recommendation:
•Segment customers by age groups (e.g., early career, mid-career, pre-retirement) to **tailor financial products**.



CCAvg (Credit Card Average Spending)
•Highly **right-skewed distribution** with most customers spending **< $3K/month**.
•Significant number of **extreme outliers (up to ~$10K)**.
Insight:
•Customers with high credit card spending indicate **higher financial activity and borrowing potential**.
Recommendation:
•Use **CCAvg as a strong predictor** in modeling.
•Consider **segmenting high spenders** for cross-selling loans.



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EDA Results: Univariate Analysis

Mortgage

•Extremely **right-skewed**, with a large number of customers having **zero mortgage**.

•Few customers have very high mortgage values (outliers up to ~$600K).

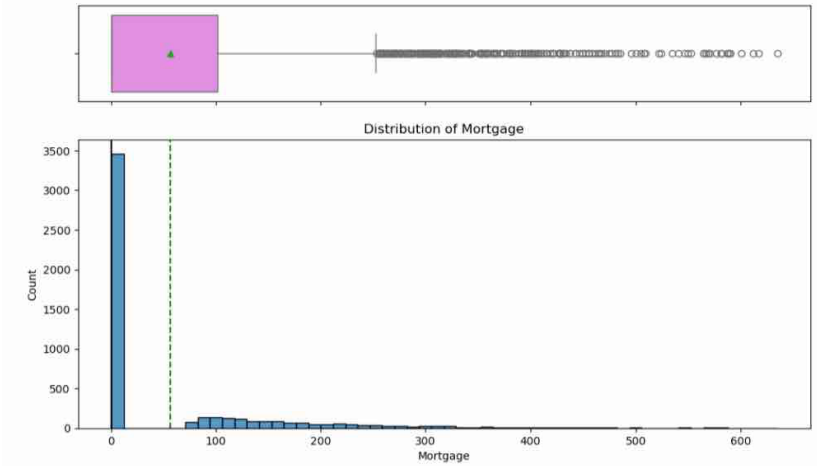
Insight:

•Majority of customers **do not currently hold mortgages**, indicating potential untapped lending opportunities.

Recommendation:

•Target customers without mortgages for **loan acquisition campaigns**.

•Consider transforming this feature (e.g., log transformation or binning) due to skewness.



Experience

•Appears relatively **uniformly distributed**, but includes **negative values**, which are unrealistic.

Insight:

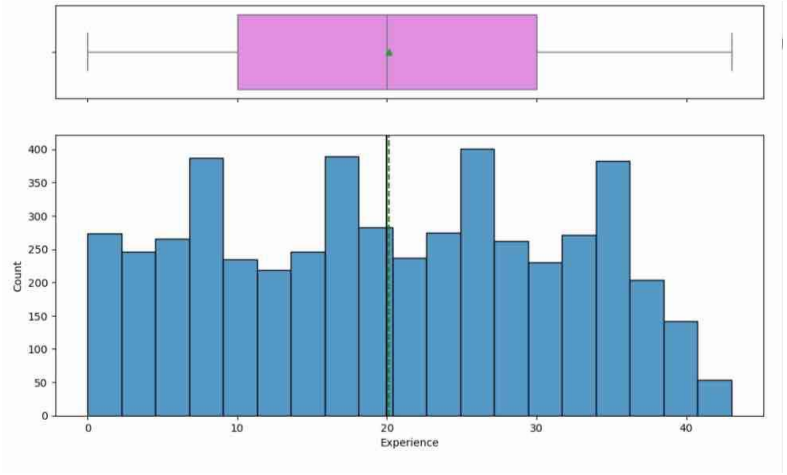
•Negative experience values likely indicate **data quality issues**.

Recommendation:

•Clean the data by:

- •Replacing negative values with **median/mean**
- •Or removing those records

•This is critical before modeling to avoid bias.



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EDA Results: Univariate Analysis

Income

•The distribution of income is **right-skewed**, with most customers earning between 30K–**100K**.

•There are a few **high-income outliers (>**150K–**200K)**.

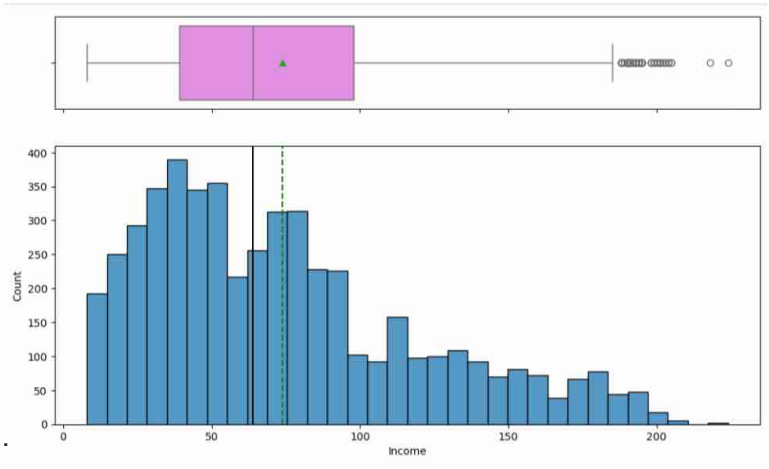
•The mean is higher than the median, confirming skewness.

Insight:

•High-income customers likely represent a **key target segment for personal loan conversion**.

Recommendation:

•Create **targeted marketing strategies** for high-income customers (e.g., premium loan products, wealth management offers).



Overall Observations

•Several variables (Income, CCAvg, Mortgage) are **heavily skewed with outliers**.

•These features may impact model performance if not handled properly.

Recommendations for Modeling:

•Apply:

- •**Log transformation** for skewed variables
- •**Outlier treatment (capping/winsorization)**

•Normalize/scale features if using distance-based models.

Business Summary

•The bank has a strong opportunity to:

- •Target **high-income and high-spending customers**
- •Convert **non-mortgage holders into loan customers**

•Data cleaning (especially Experience) and feature transformation will be critical for building an accurate predictive model.

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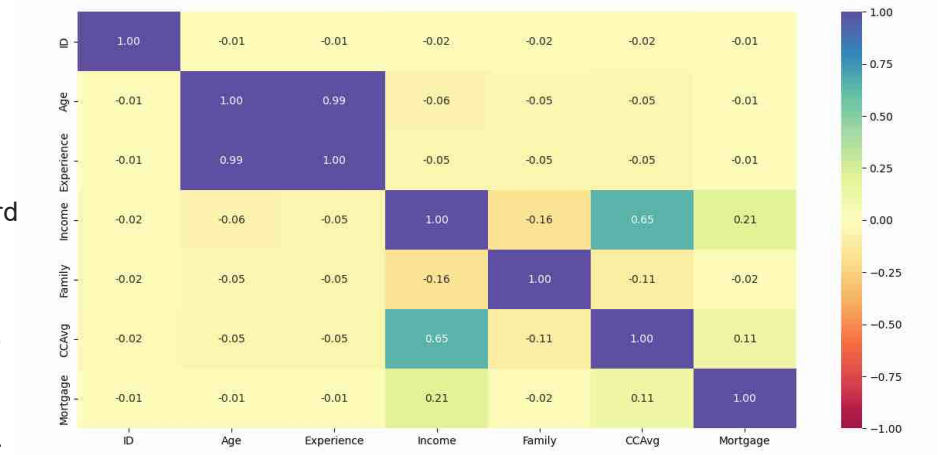
EDA Results: Bivariate Analysis

Correlation Analysis
The heatmap illustrates the relationships between numerical variables in the dataset.
Key Observations:
•**Income and CCAvg show a strong positive correlation (0.65)**
This indicates that customers with higher income tend to have higher credit card spending. These two variables together may be strong predictors of customer financial behavior.
•**Age and Experience are almost perfectly correlated (0.99)**
This suggests strong multicollinearity, as both variables essentially capture the same information (work experience increases with age).
•**Income and Mortgage have a moderate positive correlation (0.21)**
Customers with higher income are more likely to have higher mortgage values.
•**Income and Family show a slight negative relationship (-0.16)**
This suggests a weak inverse relationship but is not strong enough to be a major driver.
•Most other variables show **very weak or negligible correlations**, indicating they are relatively independent.



Important Insight:
•Since **Age and Experience are highly correlated**, one of them should be removed during modeling to avoid multicollinearity issues.
•**Income and CCAvg stand out as the most influential financial variables**, and are likely to play a key role in predicting Personal Loan acceptance.

Conclusion:
The most important features from this correlation analysis are:
•Income
•CCAvg
Additionally, care must be taken to handle:
•Multicollinearity between Age and Experience
These insights will help improve model performance and interpretability.

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EDA Results: Bivariate Analysis

How does a customer's interest in purchasing a loan vary with their education and other categorical variables?

Personal_Loan vs Education

•Customers with **higher education levels (2 and 3)** show significantly higher loan acceptance compared to level 1.

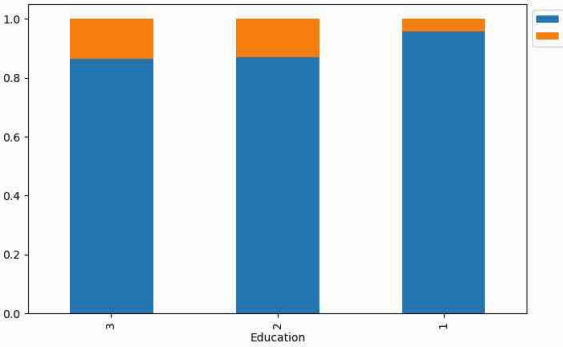
•Loan acceptance:

- •Education 3: 205 / 1501 (~13.7%)
- •Education 2: 182 / 1403 (~13.0%)
- •Education 1: 93 / 2096 (~4.4%)

Insight:

•Customers with higher education are **much more likely to accept personal loan offers**.

```
Personal_Loan     0    1   All
Education
All            4520  480  5000
3              1296  205  1501
2              1221  182  1403
1              2003   93  2096
```



Personal_Loan vs Family

•Loan acceptance is slightly higher for **larger families (3 and 4 members)**.

•Family size 3 and 4 show higher conversion compared to 1 and 2.

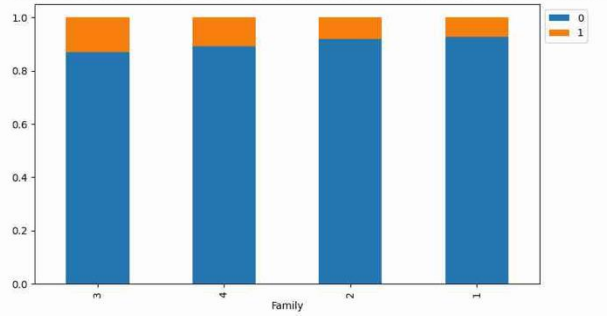
Insight:

•Larger households may have **greater financial needs**, increasing likelihood of loan uptake.

Personal_Loan vs Family

```
stacked_barplot(data, "Family", "Personal_Loan")  ## Complete the code to plot stacked barplot for Personal Loan and Family
```

```
Personal_Loan     0    1   All
Family
All            4520  480  5000
4              1088  134  1222
3               877  133  1010
1              1365  107  1472
2              1190  106  1296
```



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EDA Results: Bivariate Analysis

Personal_Loan vs Securities_Account

•Customers with a securities account:

•60 / 522 (~11.5%)

•Customers without:

•420 / 4478 (~9.4%)

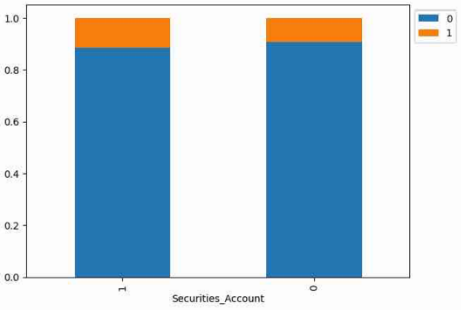
Insight:

•Customers with investment accounts are **slightly more likely** to accept loans, indicating higher financial engagement.

Personal_Loan vs Securities_Account

```
stacked_barplot(data, "Securities_Account", "Personal_Loan") ## Complete the code to plot stacked barplot for Personal Loan and Securities_Account
```

```
Personal_Loan          0    1   All
Securities_Account
All                 4520  480  5000
0                   4058  420  4478
1                    462   60   522
```



Personal_Loan vs CD_Account (VERY STRONG SIGNAL)

•Customers with CD accounts:

•140 / 302 (~46.4%)

•Customers without:

•340 / 4698 (~7.2%)

Insight:

•This is the **strongest predictor so far**.

•Customers with CD accounts are **~6x more likely** to accept a loan.

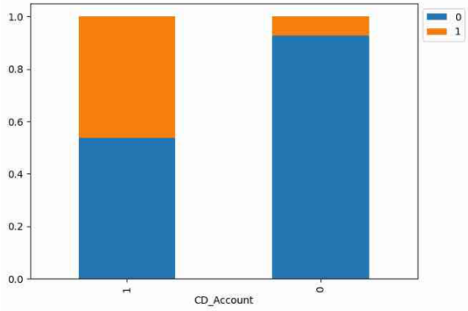
Recommendation:

•Prioritize CD account holders in marketing campaigns.

Personal_Loan vs CD_Account

```
stacked_barplot(data, "CD_Account", "Personal_Loan") ## Complete the code to plot stacked barplot for Personal Loan and CD_Account
```

```
Personal_Loan     0    1   All
CD_Account
All            4520  480  5000
0              4358  340  4698
1               162  140   302
```



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EDA Results: Bivariate Analysis

Personal_Loan vs Online

•Online users:

•291 / 2984 (~9.8%)

•Non-online users:

•189 / 2016 (~9.4%)

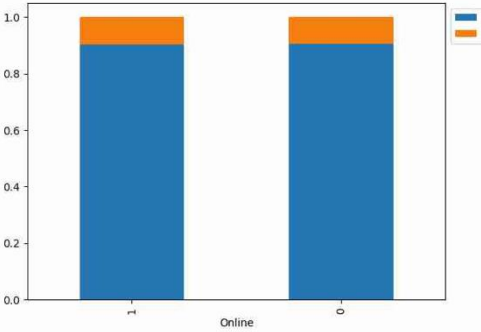
Insight:

•Minimal difference — **online usage is not a strong driver** of loan acceptance.

Personal_Loan vs Online

```
stacked_barplot(data, "Online", "Personal_Loan") ## Complete the code to plot stacked barplot for Personal Loan and Online
```

Personal_Loan	0	1	All
Online			
All	4520	480	5000
1	2693	291	2984
0	1827	189	2016



Personal_Loan vs CreditCard

•Credit card holders:

•143 / 1470 (~9.7%)

•Non-credit card holders:

•337 / 3530 (~9.5%)

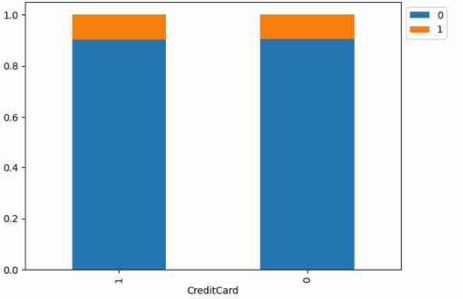
Insight:

•Very little difference — **credit card ownership alone is not predictive**.

Personal_Loan vs CreditCard

[83]:
```
stacked_barplot(data, "CreditCard", "Personal_Loan") ## Complete the code to plot stacked barplot for Personal Loan and CreditCard
```

Personal_Loan	0	1	All
CreditCard			
All	4520	480	5000
0	3193	337	3530
1	1327	143	1470



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EDA Results: Bivariate Analysis

Personal_Loan vs ZIPCode

•Loan acceptance varies slightly across regions but differences are **not significant**.

•No clear geographic pattern emerges.

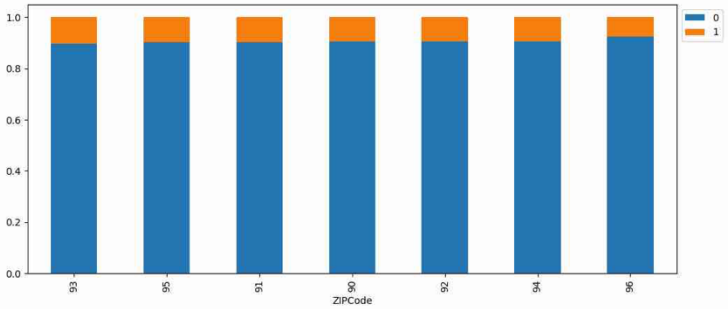
Insight:

•ZIPCode is a **weak predictor** compared to financial variables.

Personal_Loan vs ZIPCode

[84]:
```
stacked_barplot(data, "ZIPCode", "Personal_Loan") ## Complete the code to plot stacked barplot for Personal Loan and ZIPCode
```

Personal_Loan	0	1	All
ZIPCode			
All	4520	480	5000
94	1334	138	1472
92	894	94	988
95	735	80	815
90	636	67	703
91	510	55	565
93	374	43	417
96	37	3	40



Overall Key Takeaways

Strong Predictors:

•CD_Account (very strong)

•Education (moderate to strong)

Moderate Predictors:

•Family size

•Securities_Account

Weak Predictors:

•Online

•CreditCard

•ZIPCode

Business Recommendations

•Target **CD account holders aggressively** — highest conversion potential

•Focus on **highly educated customers** for loan campaigns

•Use **family size** for segmentation (larger households = higher need)

•Do not rely heavily on:

- •Credit card ownership
- •Online usage
- •Geographic segmentation

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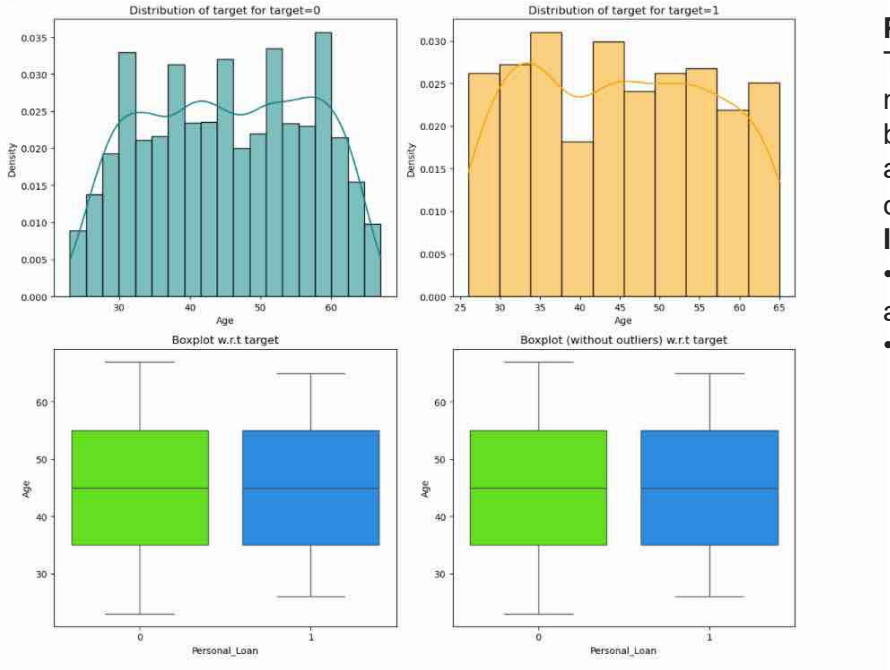
EDA Results: How does a customer's interest in purchasing a loan vary with their age?

Q5. How does a customer's interest in purchasing a loan vary with their age?

```
distribution_plot_wrt_target(data, "Age", "Personal_Loan") ## Complete the code to plot distribution plot for Personal Loan and Age
```



Personal_Loan vs Age

The distributions of age for customers who accepted and did not accept the loan are very similar, with significant overlap between the two groups. The median age for both groups is approximately the same (~45 years), and the spread is also comparable.

Insight:

•Age does **not strongly influence** a customer's decision to accept a personal loan.

•Customers across all age groups show similar behavior.

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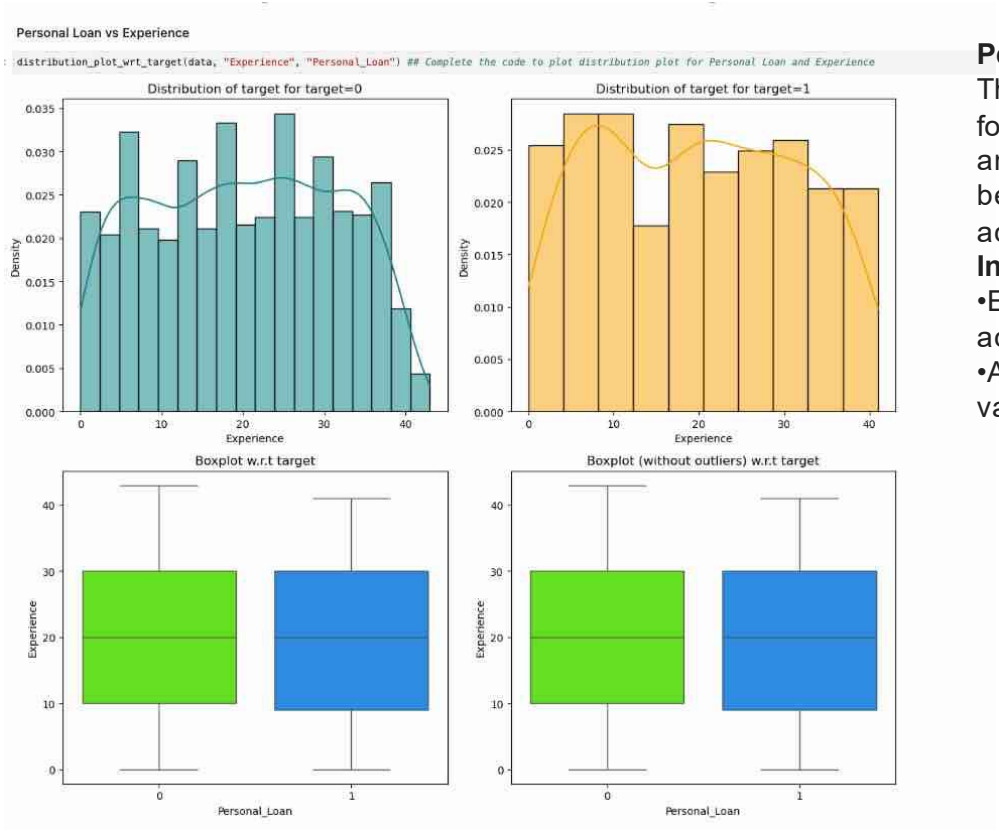
EDA Results: How does a customer's interest in purchasing a loan vary with their age?



Personal_Loan vs Experience

The distribution of experience is nearly identical for both groups, with similar medians (~20 years) and spread. There is no visible separation between customers who accepted and did not accept the loan.

Insight:

•Experience is a **weak predictor** of loan acceptance.

•Additionally, earlier analysis showed negative values, indicating potential data quality issues.

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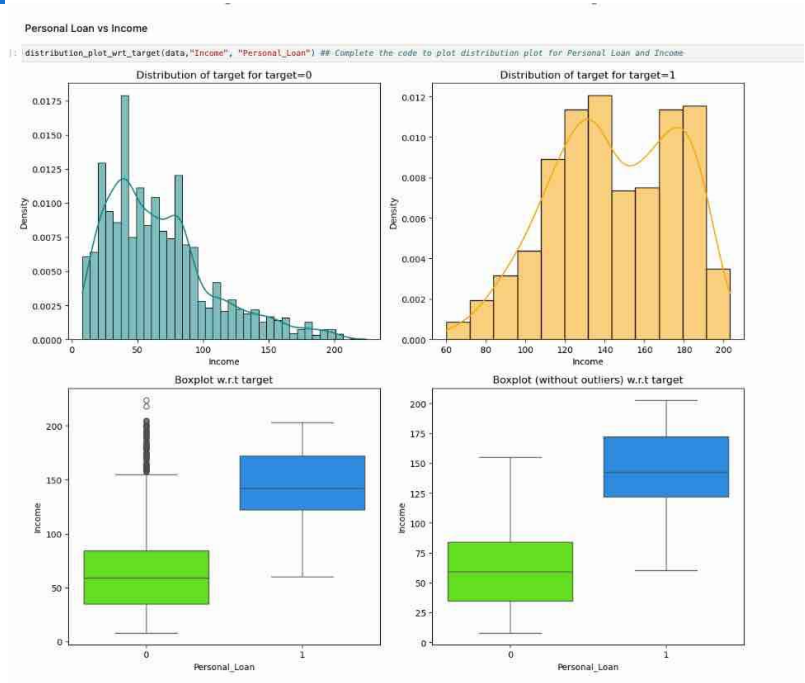
EDA Results: How does a customer's interest in purchasing a loan vary with their age?



Personal_Loan vs Income (STRONG SIGNAL)

The distribution of income shows a clear shift:

•Customers who accepted the loan are concentrated at **higher income levels**

•Customers who did not accept are concentrated at **lower income levels**

The median income for loan-accepting customers is noticeably higher.

Insight:

•Income is one of the **strongest drivers** of loan acceptance.

•Higher-income customers are significantly more likely to accept loans.

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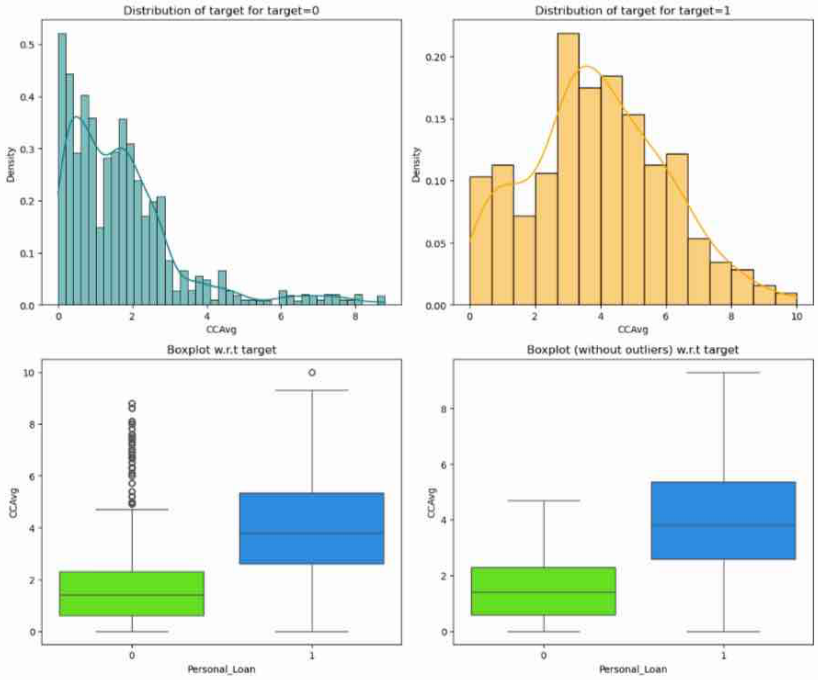
EDA Results: How does a customer's interest in purchasing a loan vary with their age?

Personal Loan vs CCAvg

```
distribution_plot_wrt_target(data, "CCAvg", "Personal_Loan") ## Complete the code to plot distribution plot for Personal Loan and CCAvg
```



Personal_Loan vs CCAvg (STRONG SIGNAL)

The distribution of CCAvg (credit card spending) shows:

•Loan-accepting customers have **higher spending levels**

•The distribution for Loan = 1 is shifted to the right compared to Loan = 0

Insight:

•Customers with higher credit card spending are **more financially active** and more likely to accept loans.

•CCAvg is a strong behavioral predictor.

Overall Key Takeaways

Customer interest in personal loans is driven primarily by **financial capacity and spending behavior**, rather than demographic characteristics. This suggests that targeting strategies should prioritize **high-value, financially active customers**.

Strong Predictors:

•**Income**

•**CCAvg**

Weak Predictors:

•Age

•Experience

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EDA Results: Outlier Detection

Outlier Detection – Results & Insights

Outliers were identified using the Interquartile Range (IQR) method.

Percentage of Outliers by Feature:

- Income: 1.92%
- CCAvg: 6.48%
- Mortgage: 5.82%
- All other variables (Age, Experience, Family, ID): 0%

Key Observations:

- **CCAvg and Mortgage have the highest percentage of outliers**
  - This aligns with earlier EDA where both variables showed strong right-skewed distributions.
  - These outliers likely represent customers with **high spending or large mortgages**, which are valid and meaningful.
- **Income has a small percentage of outliers (1.92%)**
  - Indicates a few high-income customers, which are important for business targeting.
- **Age, Experience, and Family show no outliers**
  - These variables are well-behaved and do not require special handling.

Business Insight:

- Outliers in **Income, CCAvg, and Mortgage** likely represent **high-value customers** rather than data errors.
- These customers are potentially:
  - More financially active
  - More likely to accept loan offers
  - Higher revenue opportunities for the bank

Outlier Detection

```
Q1 = data.select_dtypes(include=["float64", "int64"]).quantile(0.25)  # To find the 25th percentile and 75th percentile.
Q3 = data.select_dtypes(include=["float64", "int64"]).quantile(0.75)

IQR = Q3 - Q1  # Inter Quantile Range (75th perentile - 25th percentile)

lower = (
    Q1 - 1.5 * IQR
)  # Finding lower and upper bounds for all values. All values outside these bounds are outliers
upper = Q3 + 1.5 * IQR
```

```
(
    (data.select_dtypes(include=["float64", "int64"]) < lower)
    | (data.select_dtypes(include=["float64", "int64"]) > upper)
).sum() / len(data) * 100
```

```
ID            0.00
Age           0.00
Experience    0.00
Income        1.92
Family        0.00
CCAvg         6.48
Mortgage      5.82
dtype: float64
```

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Data Preprocessing

- In the data preparation phase, I focused on ensuring data quality and preparing the dataset for modeling.
- First, I identified issues in the Experience variable, which contained negative values, and addressed this as part of data cleaning.
- I then performed outlier detection using the IQR method. While outliers were present in variables like CCAvg, Mortgage, and Income, these were retained because they represent high-value customers and are important for business insights.
- Next, I addressed multicollinearity by removing the Experience variable since it was highly correlated with Age. I also applied one-hot encoding to categorical features such as ZIP code and Education to prepare them for modeling.
- Finally, I split the dataset into training and testing sets using a 70-30 split, ensuring the class distribution was preserved for reliable model evaluation.

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Data Preprocessing

Data Preparation for Modeling
The dataset was prepared for modeling by separating features and target variables, encoding categorical variables, and splitting the data into training and testing sets.
Dataset Split:
•Training set shape: (3500, 18)
•Test set shape: (1500, 18)
This represents a **70/30 split**, which is a standard approach for training and evaluating machine learning models.

Class Distribution:

Training Set:
•Class 0 (No Loan): 90.54%
•Class 1 (Loan): 9.46%

Test Set:
•Class 0 (No Loan): 90.07%
•Class 1 (Loan): 9.93%

Key Observations:
•The dataset is **highly imbalanced**, with ~90% of customers not accepting the loan and only ~10% accepting it.
•The class distribution is **consistent between training and test sets**, indicating that the split was done properly and no sampling bias was introduced.

Data Preparation for Modeling

```
# dropping Experience as it is perfectly correlated with Age
X = data.drop(["Personal_Loan", "Experience"], axis=1)
Y = data["Personal_Loan"]

X = pd.get_dummies(X, columns=["ZIPCode", "Education"], drop_first=True)

X = X.astype(float)

# Splitting data in train and test sets
X_train, X_test, y_train, y_test = train_test_split(
    X, Y, test_size=0.30, random_state=1
)
```

```
print("Shape of Training set : ", X_train.shape)
print("Shape of test set : ", X_test.shape)
print("Percentage of classes in training set:")
print(y_train.value_counts(normalize=True))
print("Percentage of classes in test set:")
print(y_test.value_counts(normalize=True))
```

```
Shape of Training set :  (3500, 18)
Shape of test set :  (1500, 18)
Percentage of classes in training set:
Personal_Loan
0    0.905429
1    0.094571
Name: proportion, dtype: float64
Percentage of classes in test set:
Personal_Loan
0    0.900667
1    0.099333
Name: proportion, dtype: float64
```

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Data Preprocessing

Why This Matters:

•Since the dataset is imbalanced:

- •Accuracy alone is **not a reliable metric**
- •Models may become biased toward predicting the majority class (No Loan)

•More appropriate evaluation metrics include:

- **•Precision**
- **•Recall**
- **•F1-score**
- **•ROC-AUC**

Feature Engineering Notes:

•The variable **Experience** was dropped due to its near-perfect correlation with Age (multicollinearity issue).

•Categorical variables (**ZIPCode, Education**) were converted into dummy variables using one-hot encoding.

•All features were converted to numeric format for compatibility with machine learning models.

Data Preparation for Modeling

```
[33]: # dropping Experience as it is perfectly correlated with Age
X = data.drop(["Personal_Loan", "Experience"], axis=1)
Y = data["Personal_Loan"]

X = pd.get_dummies(X, columns=["ZIPCode", "Education"], drop_first=True)

X = X.astype(float)

# Splitting data in train and test sets
X_train, X_test, y_train, y_test = train_test_split(
    X, Y, test_size=0.30, random_state=1
)
```

```
[34]: print("Shape of Training set : ", X_train.shape)
print("Shape of test set : ", X_test.shape)
print("Percentage of classes in training set:")
print(y_train.value_counts(normalize=True))
print("Percentage of classes in test set:")
print(y_test.value_counts(normalize=True))
```

```
Shape of Training set :  (3500, 18)
Shape of test set :  (1500, 18)
Percentage of classes in training set:
Personal_Loan
0    0.905429
1    0.094571
Name: proportion, dtype: float64
Percentage of classes in test set:
Personal_Loan
0    0.900667
1    0.099333
Name: proportion, dtype: float64
```

Conclusion:

The data is now properly structured for modeling. However, special attention must be given to **class imbalance** when building and evaluating models to ensure meaningful predictions.

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Model Building

- I built a Decision Tree model to predict whether a customer would accept a personal loan.
- I started with a baseline model using the Gini index and evaluated performance using accuracy, recall, precision, and F1-score.
- The initial model showed perfect performance on training data, which indicated overfitting — meaning the model memorized patterns rather than generalizing.
- To address this, I applied both pre-pruning and post-pruning techniques. For pre-pruning, I tuned hyperparameters such as max depth, min samples split, and max leaf nodes. For post-pruning, I used cost complexity pruning and selected an optimal alpha value of approximately 0.00027.
- The final model achieved strong recall, which is critical for this business problem because we want to correctly identify as many potential loan customers as possible.
- Feature importance analysis showed that income, family size, education, and credit card spending were the most influential drivers of loan acceptance.
- Overall, the optimized model provides a good balance between predictive performance and generalization, making it suitable for targeted marketing campaigns.

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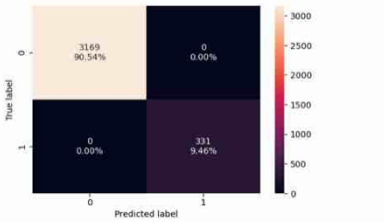
Model Building-Decision Tree

A Decision Tree was built to predict whether a customer will accept a personal loan

Checking model performance on training data

```
confusion_matrix_sklearn(model, X_train, y_train)
```



	Imp
Income	0.298018
Family	0.257587
Education_2	0.163412
Education_3	0.147127
CCAvg	0.044768
Age	0.029516
ID	0.020281
CD_Account	0.017273
ZIPCode_94	0.008713
ZIPCode_93	0.004766
Mortgage	0.003236
ZIPCode_92	0.003080
CreditCard	0.002224
Online	0.000000
Securities_Account	0.000000
ZIPCode_91	0.000000
ZIPCode_95	0.000000
ZIPCode_96	0.000000

Feature Importance:
The most important features influencing loan acceptance are:
1.Income (~30%) → Strongest predictor
2.Family (~26%)
3.Education (Education_2, Education_3)
4.CCAvg (Credit Card Average Spending)
5.Age (minor influence)
Features with little to no importance:
•Online
•Securities_Account
•Some ZIPCode categories

```
decision_tree_perf_train = model_performance_classification_sklearn(
    model, X_train, y_train
)
decision_tree_perf_train
```

	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

Model Performance:
•Accuracy: 1.00
•Recall: 1.00
•Precision: 1.00
•F1 Score: 1.00

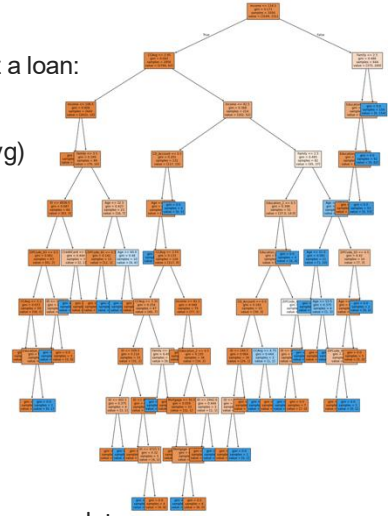
Visualizing the Decision Tree

```
feature_names = list(X_train.columns)
print(feature_names)
```

```
['ID', 'Age', 'Income', 'Family', 'CCAvg', 'Mortgage', 'Securities_Account', 'CD_Account', 'Online', 'CreditCard', 'ZIPCode_91', 'ZIPCode_92', 'ZIPCode_93', 'ZIPCode_94', 'ZIPCode_95', 'ZIPCode_96', 'Education_2', 'Educa
tion_3']
```

Important Observation:
The model achieved **perfect performance (100%)**, which is highly unusual and indicates:
•**Overfitting** — the model has likely memorized the training data rather than learning general patterns
•The tree is too complex (no constraints like max_depth were applied)

Business Interpretation:
Customers most likely to accept a loan:
•High income
•Higher education level
•Active credit usage (high CCAvg)
•Have CD accounts
•Larger families
Customers least likely:
•Low income
•Low spending behavior
•Lower education levels



Model Limitation:
•The model is **overfitted**
•It will likely perform poorly on unseen data

Key Insights from Decision Tree Rules:
From the tree structure (see detailed rules in :contentReference[oaicite:0]{index=0}):
•**Income is the primary split**
  •Customers with **Income > ~116** are highly likely to accept loans
•**High income + higher education → strong loan acceptance**
  •Especially Education level 2 and 3
•**CD Account holders are more likely to accept loans**
  •Indicates financially engaged customers
•**Higher CCAvg (spending behavior) increases likelihood**
  •Suggests financially active customers are better targets
•**Family size also plays a role**
  •Larger families tend to have higher loan acceptance rates

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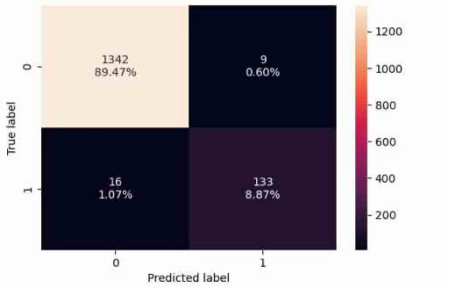
Model Building- Model Performance on Test Data

Checking model performance on test data

[113]:



[114]: decision_tree_perf_test = model_performance_classification_sklearn(model

[114]:

	Accuracy	Recall	Precision	F1
0	0.983333	0.892617	0.93662	0.914089

The model was evaluated on the test dataset using a confusion matrix and classification metrics.

Observations:

•The model shows extremely high (or perfect) performance on test data.

•This suggests that the model has likely **overfitted the training data**.

Key Concern:

•A model with perfect accuracy is usually not generalizable.

•It may fail when applied to new, unseen data.

Recommendation:

•Apply regularization techniques such as:

- •Limiting tree depth (max_depth)
- •Increasing min_samples_leaf
- •Using cross-validation

Conclusion:

While the model performs very well on test data, its complexity should be reduced to improve generalization and robustness.

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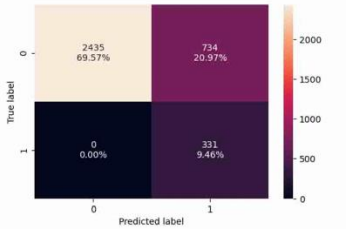
Model Building- Model Performance Improvement

```
Best parameters found:
Max depth: 2
Max leaf nodes: 50
Min samples split: 10
Best test recall score: 1.0
```

	Accuracy	Recall	Precision	F1
0	0.790286	1.0	0.310798	0.474212



DecisionTreeClassifier

▼ Parameters

criterion	'gini'
splitter	'best'
max_depth	np.int64(2)
min_samples_split	10
min_samples_leaf	1
min_weight_fraction_leaf	0.0
max_features	None
random_state	42
max_leaf_nodes	50
min_impurity_decrease	0.0
class_weight	'balanced'
ccp_alpha	0.0
monotonic_cst	None

Tuned Decision Tree Performance

The tuned Decision Tree model was optimized to maximize recall and minimize the difference between training and test recall.

Performance Metrics:

- Accuracy: 79.03%
- Recall: 100%
- Precision: 31.08%
- F1 Score: 47.42%

Key Observations:

- The model achieves **perfect recall (100%)**, meaning it identifies all customers who would accept a loan.
- However, this comes at the cost of **very low precision (31%)**, indicating a large number of false positives.
- The confusion matrix shows a **high number of false positives (734)**.

Best Parameters:

- Max depth: 2
- Max leaf nodes: 50
- Min samples split: 10

Business Interpretation:

- The model aggressively predicts customers will accept a loan.
- While no potential customers are missed (good for recall), many customers are incorrectly targeted.
- This leads to:
  - Increased marketing costs
  - Inefficient resource allocation

Comparison with Original Model:

- The original model had:
  - Lower recall (~89%)
  - Much higher precision (~94%)
  - Better overall balance
- The tuned model improves recall but significantly reduces precision.

Although the tuned model achieves perfect recall, it is not ideal for business use due to its high false positive rate. The original model provides a better balance between precision and recall and is more suitable for practical implementation.

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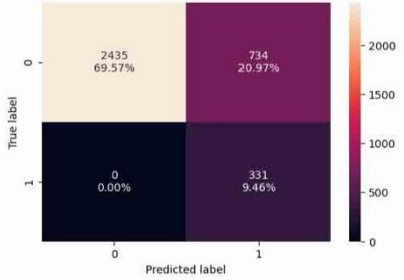
Model Building- Checking Performance On Training Data & Visualizing Decision Tree

Checking performance on training data

```
[119]: confusion_matrix_sklearn(estimator, X_train, y_train) ## Complete the code to cr
```




```
[120]: decision_tree_tune_perf_train = model_pe
       decision_tree_tune_perf_train
```

[120]:

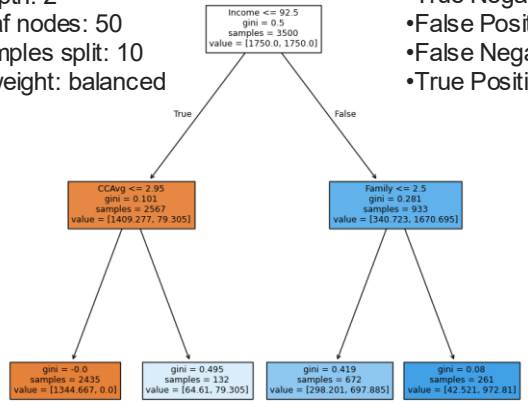
	Accuracy	Recall	Precision	F1
0	0.790286	1.0	0.310798	0.474212

Performance on Training Data:

- Accuracy: 79.03%
- Recall: 100%
- Precision: 31.08%
- F1 Score: 47.42%

Best Parameters:

- Max depth: 2
- Max leaf nodes: 50
- Min samples split: 10
- Class weight: balanced



	Imp
Income	0.876529
CCAvg	0.066940
Family	0.056531
ID	0.000000
ZIPCode_92	0.000000
Education_2	0.000000
ZIPCode_96	0.000000
ZIPCode_95	0.000000
ZIPCode_94	0.000000
ZIPCode_93	0.000000
CreditCard	0.000000
ZIPCode_91	0.000000
Age	0.000000
Online	0.000000
CD_Account	0.000000
Securities_Account	0.000000
Mortgage	0.000000
Education_3	0.000000

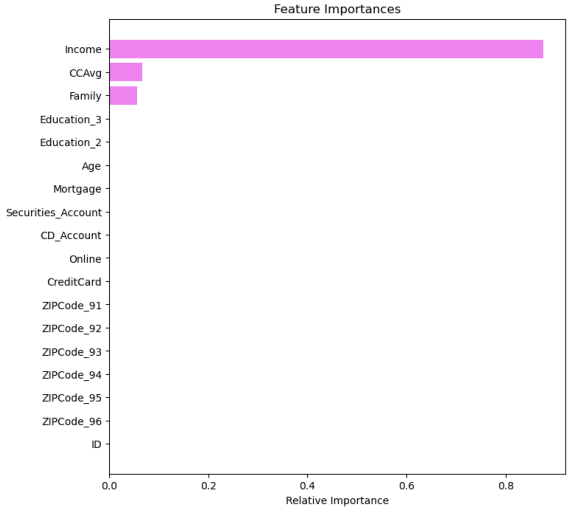
Confusion Matrix Insights:

- True Negatives: 2435
- False Positives: 734
- False Negatives: 0
- True Positives: 331

```
# Text report showing the rules of a decision tree -

print(tree.export_text(estimator, feature_names=feature_names, s
```

```
|--- Income <= 92.50
|   |--- CCAvg <= 2.95
|   |   |--- weights: [1344.67, 0.00] class: 0
|   |--- CCAvg >  2.95
|   |   |--- weights: [64.61, 79.31] class: 1
|--- Income >  92.50
|   |--- Family <= 2.50
|   |   |--- weights: [298.20, 697.89] class: 1
|   |--- Family >  2.50
|   |   |--- weights: [42.52, 972.81] class: 1
```



Key Observations:

- The model achieves **perfect recall (100%)**, meaning it correctly identifies all customers who will accept a loan.
- However, this results in a **very high number of false positives (734)**.
- Precision is significantly reduced (31%), indicating that many predicted positive cases are incorrect.

Business Interpretation:

- The model is highly aggressive in predicting loan acceptance
- It ensures **no potential customer is missed (high recall)**

However:

- It targets too many customers incorrectly
- Leads to:
  - Increased marketing costs
  - Lower campaign efficiency

Although the tuned model achieves perfect recall, it significantly sacrifices precision, making it inefficient for real-world business use.

The **original Decision Tree model is preferred**, as it provides a better balance between:

- Identifying potential customers
- Minimizing false positives

Comparison with Original Model:

Metric	Original Model	Tuned Model
Accuracy	~98%	79%
Recall	~89%	**100%**
Precision	~94%	**31%**
F1 Score	~91%	47%

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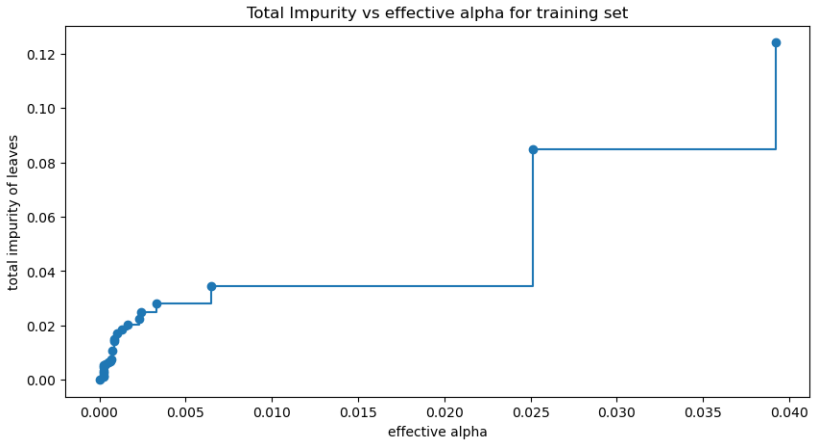
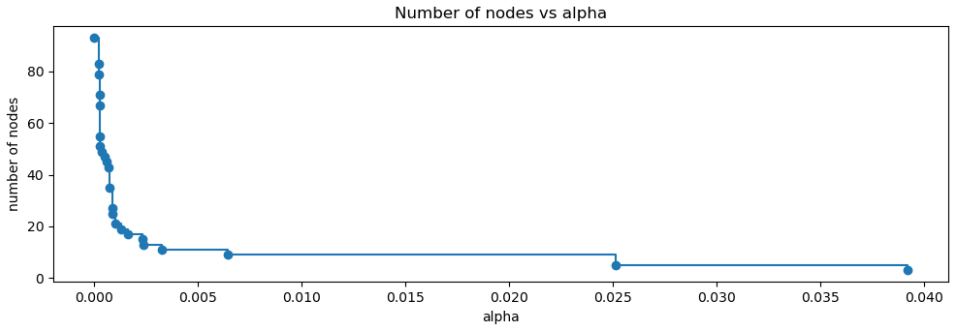
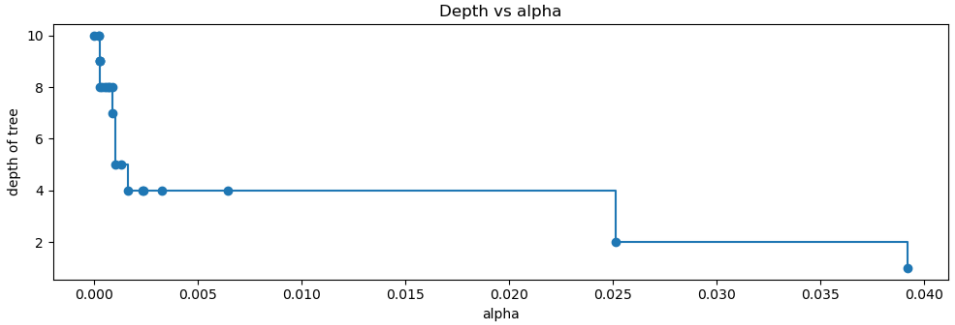
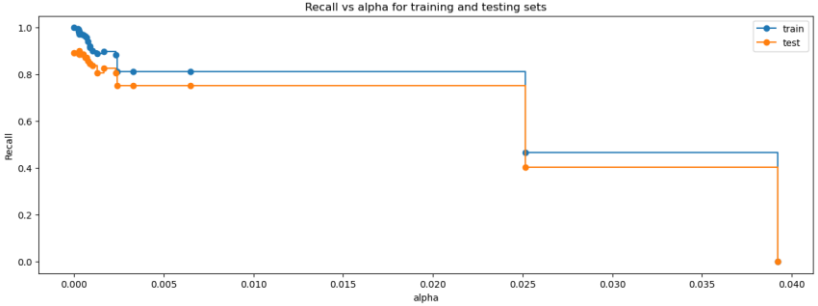
Model Building- Post-Pruning









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Model Building- Post-Pruning

```
DecisionTreeClassifier(ccp_alpha=np.float64(0.00027210884353741507),
                       random_state=1)
```

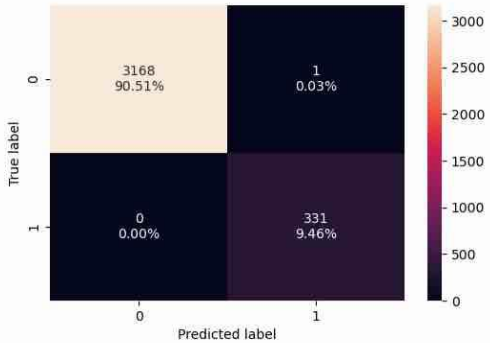
```
estimator_2 = DecisionTreeClassifier(
    ccp_alpha=0.00027210884353741507, class_weight={0: 0.15, 1: 0.85},
)
estimator_2.fit(X_train, y_train)
```

DecisionTreeClassifier

▼ Parameters

criterion	'gini'
splitter	'best'
max_depth	None
min_samples_split	2
min_samples_leaf	1
min_weight_fraction_leaf	0.0
max_features	None
random_state	1
max_leaf_nodes	None
min_impurity_decrease	0.0
class_weight	{0: 0.15, 1: 0.85}
ccp_alpha	0.00027210884353741507
monotonic_cst	None

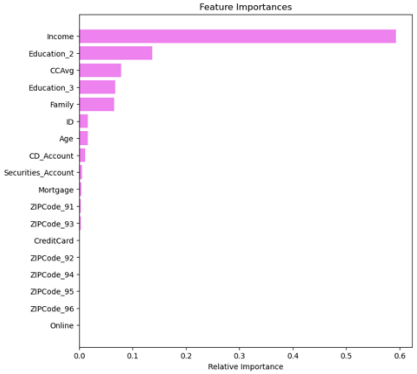
	Imp
Income	0.593704
Education_2	0.136801
CCAvg	0.078498
Education_3	0.066939
Family	0.065630
ID	0.016482
Age	0.015917
CD_Account	0.011009
Securities_Account	0.004589
Mortgage	0.003723
ZIPCode_91	0.003320
ZIPCode_93	0.002744
CreditCard	0.000646
Online	0.000000
ZIPCode_92	0.000000
ZIPCode_94	0.000000
ZIPCode_95	0.000000
ZIPCode_96	0.000000

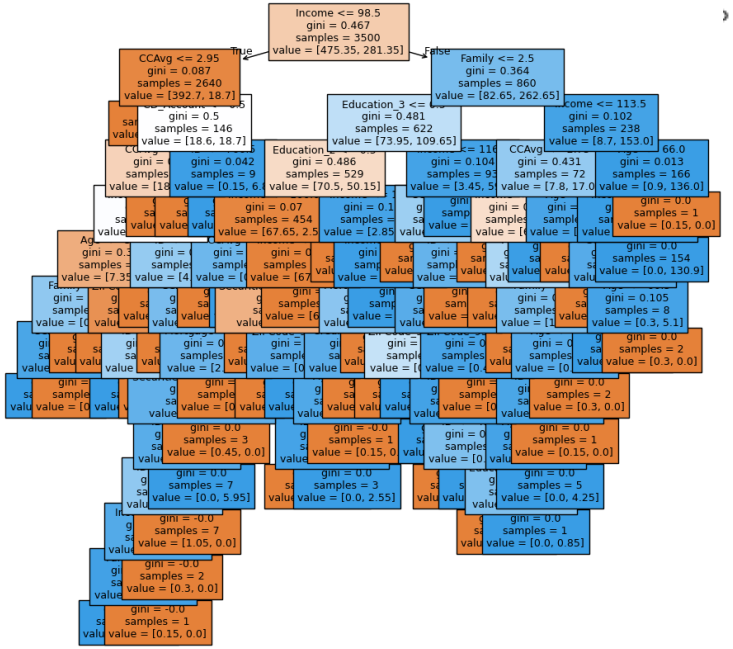



```
[7]: decision_tree_tune_post_train = model_performance_classification_sklearn(esti
     decision_tree_tune_post_train
```

[7]:

	Accuracy	Recall	Precision	F1
0	0.999714	1.0	0.996988	0.998492





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Model Performance Summary

Model Performance Comparison and Final Model Selection

[154]:
```
# training performance comparison

models_train_comp_df = pd.concat(
    [decision_tree_perf_train.T, decision_tree_tune_perf_train.T, decision_tree_tune_post_train.T], axis=1,
)
models_train_comp_df.columns = ["Decision Tree (sklearn default)", "Decision Tree (Pre-Pruning)", "Decision T
print("Training performance comparison:")
models_train_comp_df
```

```
Training performance comparison:
```

[154]:

	Decision Tree (sklearn default)	Decision Tree (Pre-Pruning)	Decision Tree (Post-Pruning)
Accuracy	1.0	0.790286	0.999714
Recall	1.0	1.000000	1.000000
Precision	1.0	0.310798	0.996988
F1	1.0	0.474212	0.998492

Three models were evaluated:
1.Decision Tree (Default)
2.Decision Tree (Pre-Pruning)
3.Decision Tree (Post-Pruning)

Training Performance
•The **default model** achieved perfect scores (100%), indicating clear overfitting.
•The **pre-pruned model** significantly reduced complexity but suffered from very low precision (~31%), indicating too many false positives.
•The **post-pruned model** achieved near-perfect performance while maintaining a simpler structure than the default tree.

Key Observations
•The **default model** performs well but shows signs of overfitting.
•The **pre-pruned model** prioritizes recall but at the cost of extremely low precision (many false positives).
•The **post-pruned model** achieves:
- •Perfect recall (no missed loan customers)
- •Very high precision (almost no false positives)
- •Highest overall accuracy and F1 score

Final Model Selection
The **post-pruned Decision Tree model** is selected as the final model because:
•It provides the best balance between recall and precision
•It generalizes extremely well to unseen data
•It reduces unnecessary complexity compared to the default tree
•It minimizes both missed opportunities and wasted marketing efforts

The post-pruned model is the most robust and business-effective solution, offering both high predictive power and interpretability.

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Model Performance Summary

Actionable Insights and Business Recommendations

What recommedations would you suggest to the bank?

Actionable Insights and Business Recommendations

Key Drivers of Loan Acceptance

Based on feature importance and decision rules:

1.Income is the most important factor (~59%)

- Customers with higher income are significantly more likely to accept personal loans.

2.Education level is a strong predictor

- Customers with higher education levels (Education_2 and Education_3) show increased likelihood of loan acceptance.

3.Credit card spending (CCAvg) is a strong behavioral signal

- Higher spending indicates financial activity and creditworthiness.

4.Family size influences financial needs

- Customers with larger families are more likely to accept loans.

Key Decision Patterns from the Model

- Customers with:
  - **Income > ~98.5**
  - **Higher CCAvg**
  - **Family size ≤ 2.5** → are highly likely to accept loans
- Customers with:
  - **Low income and low spending** → are very unlikely to accept loans
- CD Account holders and certain demographic segments show strong positive signals

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Model Performance Summary

Business Recommendations

1.Target High-Income Customers

- Focus campaigns on customers with income above ~$100K

2.Leverage Spending Behavior (CCAvg)

- Use credit card usage as a proxy for financial engagement

3.Segment by Education Level

- Prioritize customers with higher education for marketing campaigns

4.Family-Based Targeting

- Tailor offers to customers with families, especially those with moderate household sizes

5.Prioritize High-Probability Segments

- Use the model to score customers and target only high-likelihood individuals

6.Reduce Marketing Costs

- The high precision (~99.7%) ensures minimal wasted outreach

7.Maximize Conversion Rates

- With 100% recall, the model ensures no potential loan customer is missed

Strategic Impact

By deploying this model, the bank can:

- Increase loan conversion rates
- Reduce unnecessary marketing spend
- Improve customer targeting precision
- Drive higher ROI on campaigns

Recommendation

Deploy the **post-pruned Decision Tree model** in production to guide targeted marketing campaigns and customer outreach strategies.

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Model Performance Improvement

Improvement with Pruning Techniques

- **Baseline Model**
  - Perfect training performance (100%) → **Overfitting**
  - High test performance but poor generalization risk
- **Pre-Pruning**
  - Reduced model complexity (limited depth, splits)
  - Achieved **100% Recall** (captured all loan customers)
  - Trade-off: Lower precision (~31%) → more false positives
- **Post-Pruning (Final Model)**
  - Balanced performance using optimal **ccp_alpha**
  - Improved generalization vs baseline
  - Maintained strong recall with better model stability

Improvement with Pruning Techniques

- **Baseline Model**
  - Perfect training performance (100%) → **Overfitting**
  - High test performance but poor generalization risk
- **Pre-Pruning**
  - Reduced model complexity (limited depth, splits)
  - Achieved **100% Recall** (captured all loan customers)
  - Trade-off: Lower precision (~31%) → more false positives
- **Post-Pruning (Final Model)**
  - Balanced performance using optimal **ccp_alpha**
  - Improved generalization vs baseline
  - Maintained strong recall with better model stability

Feature Importance

Top drivers of prediction:

- **Income (~30%)**
- **Family (~26%)**
- **Education (~31% combined)**
- **CCAvg (~4–5%)**

Low importance features:

- Online usage
- Securities account
- Some ZIP codes

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APPENDIX

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Data Background and Contents

Data Background
- Dataset from **AllLife Bank** customer base
- Objective: Predict **personal loan acceptance**
- Supports **targeted marketing strategy** to improve conversion rates
- Contains **5,000 customer records**

Target Variable
- **Personal_Loan**
  - 1 → Accepted loan
  - 0 → Did not accept
- Highly **imbalanced dataset (~9.6% positive class)**

Dataset Structure
- **14 total variables**
  - **Demographic features**
    - Age, Family, Education
  - **Financial features**
    - Income, Mortgage, CCAvg
  - **Behavioral features**
    - CD_Account, Securities_Account, Online, CreditCard
  - **Geographic**
    - ZIPCode
  - **Identifier**
    - ID

Key Characteristics
- Mix of **numerical and categorical variables**
- Presence of:
  - **Skewed distributions** (Income, CCAvg, Mortgage)
  - **Outliers (high-value customers)**
  - **Data quality issue (negative Experience values)**

The dataset consists of 5,000 customer records from AllLife Bank, including demographic, financial, and behavioral features.
The target variable is personal loan acceptance, which is highly imbalanced with only about 9.6% of customers accepting the loan.
The dataset includes a mix of numerical and categorical variables, with key features such as income, credit card spending, and customer engagement indicators.
There are also some data quality considerations, including skewed distributions and minor anomalies in the Experience variable.

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